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Variability in first impressions of autistic adults made by neurotypical raters is driven more by characteristics of the rater than by characteristics of autistic adults

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Abstract

Previous work indicates that first impressions of autistic adults are more favorable when neurotypical raters know their clinical diagnosis and have high understanding about autism, suggesting that social experiences of autistic adults are affected by the knowledge and beliefs of the neurotypical individuals they encounter. Here, we examine these patterns in more detail by assessing variability in first impression ratings of autistic adults ($N = 20$) by neurotypical raters ($N = 505$). Variability in ratings was driven more by characteristics of raters than those of autistic adults, particularly for items related to "intentions to interact." Specifically, variability in rater stigma toward autism and autism knowledge contributed to first impression ratings. Only ratings of "awkwardness" were driven more by characteristics of the autistic adults than characteristics of the raters. Furthermore, although first impressions of autistic adults generally improved when raters were informed of their autism status, providing a diagnosis worsened impressions made by neurotypical raters with high stigma toward autism. Variations in how the diagnosis was labeled (e.g. "autistic" vs "has autism") did not affect results. These findings indicate a large role of neurotypical perceptions and biases in shaping the social experiences for autistic adults that may be improved by reducing stigma and increasing acceptance.

Keywords

adults, autism spectrum disorder, diagnostic disclosure, first impressions, stigma

Introduction

First impressions are formed rapidly and are highly resistant to change (Ambady, Bernieri, & Richeson, 2000). Favorable first impressions increase the likelihood and quality of subsequent interaction and can affect outcomes across a wide range of social contexts, including job interviews, dating, and friendship formation (Harris & Garris, 2008). For autistic people,[1] social presentations and behavior can differ from those of neurotypical (NT) individuals (Faso, Sasson, & Pinkham, 2015; Hubbard, Faso, Assmann, & Sasson, 2017; Morrison et al., 2017), and these differences can affect the formation of first impressions. Recent work has focused on how autistic adults are perceived by NT individuals, with many studies finding that NTs not only form less favorable first impressions of autistic adults and adolescents, but also report greater reluctance to socially engage with them (Grossman, 2015; Sasson et al., 2017; Sasson & Morrison, 2019). Such impressions can serve as a barrier to inclusion, limit social opportunities, and present challenges for achieving personal and professional goals (Cage, Di Monaco, & Newell, 2018).

Although autistic adolescents and adults are reliably rated by NT peers as more awkward, less likable, and less attractive than NT controls (Grossman, 2015; Sasson et al., 2017), recent evidence suggests that impressions improve when NT observers are aware they are evaluating an autistic person (Matthews, Ly, & Goldberg, 2015; Sasson & Morrison, 2019) and when they themselves are more knowledgeable about autism (Gillespie-Lynch et al., 2015; Sasson & Morrison, 2019; Tipton & Blacher, 2014; White, Hillier, Frye, & et al., 2016). Similarly, acceptance of

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autistic peers is also higher among NTs who have previous experience with autism, and this acceptance is predictive of future intentions to interact with autistic individuals (Gardiner & Iarocci, 2014). These findings suggest that perceptions of—and responses to—autistic adults are not driven solely by differences in their social presentation and expressivity (Faso et al., 2015; Hubbard et al., 2017; Morrison et al., 2017), but also by characteristics of NT perceivers.

This study examines the role of the rater in forming first impressions of autistic adults to assess how much impression formation is accounted for by the autistic adults themselves relative to characteristics of the NT raters. A finding that variability in impression ratings is accounted more by characteristics of autistic adults than by characteristics of the NT raters would suggest a consistency in NT impression formation driven primarily by reliably perceived differences (e.g. prosodic qualities of speech and emotional expressivity; Faso et al., 2015; Hubbard et al., 2017) between autistic adults. Previous work has found that some characteristics of autistic adults, such as their autistic traits or whether they disclose their diagnosis, do affect first impression ratings made by NT observers (Sasson & Morrison, 2019). NTs rate autistic individuals more positively when made aware of their clinical status, possibly because this knowledge provides an explanation for perceived atypicalities in social presentation (Matthews et al., 2015; Sasson & Morrison, 2019).

However, if results show that a large portion of the variability in first impression ratings of autistic adults is driven by characteristics of the NT raters, this would suggest that impressions formed of autistic adults are not wholly determined by the autistic adults themselves, but rather are influenced by the beliefs, biases, and expectations NT observers bring to their evaluative judgments. Indeed, recent work suggests that just as autism spectrum disorder (ASD) adults often struggle to interpret the mental states of NTs (Baron-Cohen, 2000), NT adults frequently have difficulty decoding the perspectives and emotions of autistic adults (Edey et al., 2016; Sheppard, Pillai, Wong, Ropar, & Mitchell, 2016). This "double empathy" problem (Milton, 2012), in which differences in social norms, expression, and expectations can lead to misunderstanding between autistic and NT individuals (Gernsbacher, 2006; Heasman & Gillespie, 2018; Jaswal & Akhtar, 2018; Milton, Heasman, & Sheppard, 2018), may affect NT interpretation of autistic adults and contribute to less favorable first impressions.

However, NT raters often differ in their first impressions of autistic adults, and it remains unclear what accounts for these differences. Whereas demographic characteristics (e.g. age and sex), personality traits, and level of autistic traits of NT raters do not predict first impressions of autistic adults (Gardiner & Iarocci, 2014; Sasson et al., 2017; Sasson & Morrison, 2019), recent work suggests that experience and knowledge of autism may account for significant variability among NTs in their first impressions, attitudes, and acceptance of autistic individuals (Gardiner & Iarocci, 2014; Gillespie-Lynch et al., 2015; Sasson & Morrison, 2019; Tipton & Blacher, 2014; White et al., 2016). Here, we assess, in particular, whether individual differences among NT raters in stigma about autism, autism knowledge, and personal experience with autism account for differences in how autistic adults are evaluated.

This study also explores how NT perceptions of autistic adults are affected by awareness of their diagnostic status. Although diagnostic awareness is related to more favorable impressions by NTs (Matthews et al., 2015; Sasson & Morrison, 2019), it is unclear whether this pattern extends to when NTs harbor more negative feelings about autism. It is also unclear whether diagnostic disclosure affects impression formation differently depending upon NTs' previous level of contact with autistic individuals (Gardiner & Iarocci, 2014). The assessment of autism stigma, knowledge, and contact allow us here to probe these patterns more deeply.

In addition, it is unclear whether diagnostic awareness of autism uniformly improves impressions made by NTs, or whether variations in diagnostic description (e.g. "is autistic" vs "on the spectrum" vs "has Asperger's") affect NT impression formation. Different diagnostic terminology can evoke distinct inferences about ability and disability, with euphemistic terms like "special needs" often resulting in more negative assumptions (Gernsbacher, Raimond, Balinghasay, & Boston, 2016). The use of different terminology indicating autism status may elicit different assumptions, biases, and beliefs in NT perceivers that in turn may affect impression formation, particularly for those who express stigma toward autism. It may be the case, for instance, that certain autism descriptors convey more disability and/or stigma to evaluators than others, affecting perceptions and first impressions. We examine here whether NT impressions of autistic adults are affected by differences in diagnostic terminology and whether these patterns vary depending upon NT autism knowledge, stigma, and contact. Taken together, understanding the effect of various descriptors and the role of NT rater beliefs, knowledge, and experience of autism can inform the degree to which impressions are driven by self-descriptive choices by autistic adults relative to characteristics of NT perceivers not under their control.

Methods

Participants

Stimulus participants who completed video recordings were 20 autistic adults (17 male, 19 Caucasian) ranging in

age from 18 to 41 (mean age = 24.50, *SD* = 5.27) with measured full-scale intelligence quotients (IQs) on the Wechsler Abbreviated Scale of Intelligence (WASI) in the normal range (86–126; mean IQ = 106.40, *SD* = 13.83). Diagnoses were confirmed using the Autism Diagnostic Observation Schedule (ADOS) or ADOS-2 (ADOS; Lord et al., 2012; Lord, Rutter, Pamela, & Dilavore Risi, 2008), depending upon when they were assessed. Raters were 505 undergraduates (mean age = 21.37, *SD* = 4.98; range: 18–52; 76% female) who participated to earn course credit for a psychology class. In total, 19 raters were excluded for technical difficulties, having an autism diagnosis, or being younger than 18. IQ estimated by the reading subtest of the Wide Range Achievement Test-3 (WRAT-3) (Wilkinson, 1993) was 108.04 (*SD* = 8.82) for the raters. All participants provided written informed consent, and the university institutional review board (IRB) approved the study.

Procedures

The video recordings of the 20 autistic adults have been used and described previously in prior studies (Sasson et al., 2017; Sasson & Morrison, 2019). Each of the 20 autistic stimulus participants were recorded as they participated in the "High Risk Social Challenge" (Gibson, Penn, Prinstein, Perkins, & Belger, 2010), a 1-min mock audition for a reality/game show. This task was chosen because it offers a compromise between experimental control and ecological validity. The first 10 s of the audition following content-free introductions (e.g. "Hi, this is my audition tape") serve as the video stimuli later rated by NT observers. No videos included any reference to diagnostic status, and individuals' names were edited out of all videos.

Rater participants later evaluated the 20 stimulus videos one at a time in a randomized order. They were randomly assigned to a label condition in which half of the videos were displayed alongside one of the following five autism descriptors: (1) "This person is autistic," (2) "This person has autism," (3) "This person has Asperger's," (4) "This person is on the autism spectrum," and (5) "This person has a disability," used as general comparison not referencing autism. The remaining videos were presented with a "This person has no diagnosis" control label. This between-subjects design prevented carryover effects that could have occurred if participants viewed the same videos multiple times with different labels. Two versions of each descriptor condition were created for counterbalancing purposes. Each alternated which videos received a descriptor and which received the "no diagnosis" control. Raters were randomly assigned to each condition (autism $n = 103$, autistic $n = 103$, on the autism spectrum $n = 103$, Asperger's $n = 101$, disability $n = 95$) and raters did not differ on race ($p = 0.19$), gender ($p = 0.26$), verbal IQ ($p = 0.86$), or age ($p = 0.19$) across the versions.

Raters were told they would be watching videos of people, some of whom had clinical diagnoses, and to complete the First Impression Scale after each video. Afterwards, they completed additional measures presented in randomized order to each participant.

Measures

First Impressions Scale (Sasson et al., 2017) has 10 items assessing the rater's impressions of stimulus participants. Of which, 6 items are of traits (awkwardness, attractiveness, trustworthiness, intelligence, assertiveness, and likability), and 4 items assess interest in interacting with the stimulus participant (e.g. "I would hang out with this person in my free time"; see Sasson et al., 2017). Participants responded using a 4-point Likert-type scale ranging from strongly disagree (1) to strongly agree (4) (Sasson & Morrison, 2019), and all items were scored such that higher scores indicated more favorable first impressions (e.g. awkwardness was reversed scored such that higher scores indicated less awkwardness). The internal consistency of the scale was good in this study (Cronbach's alpha = 0.76), and each trait was assessed as a separate outcome in analyses.

Autism Awareness Scale (Gillespie-Lynch et al., 2015) is a 13-item self-report questionnaire measuring participants' knowledge of autism. Participants responded to each statement using a 5-point Likert-type scale ranging from strongly disagree (–2) to strongly agree (2), yielding a score ranging from −26 to 26, with higher values indicating more knowledge of autism. This measure was validated as having strong reliability (Cronbach's alpha = 0.87; Gillespie-Lynch et al., 2015), but in the current sample this measure showed only fair reliability (Cronbach's alpha = 0.68).

Social Distance Scale (Gillespie-Lynch et al., 2015) assesses self-reported stigma toward autistic individuals using a 4-point scale ranging from *definitely unwilling* to *definitely willing* in six different situations (e.g. "How willing would you be to have a person with autism marry into the family"). Items were summed to yield a social distance score ranging from 6 to 24, where higher values indicated more social distance and stigma toward autism. The reliability of the scale in this study was good (Cronbach's alpha = 0.86).

Level of Contact Report (Gardiner & Iarocci, 2014; Holmes, Corrigan, Williams, Canar, & Kubiak, 1999). Participants read 12 statements describing a relationship with an autistic person or an experience with autistic people. Each of the items has a rank order score indicating intensity of relations with autism. This ranges from 1 ("I have never observed a person that I was aware had autism spectrum disorder") to 12 ("I have an autism spectrum disorder"). Participants were allowed to check as many relationships as they had with autism, and their final score was his or her highest level of contact endorsed.

Data analytic plan

Multilevel modeling (MLM) with restricted maximum likelihood (REML) estimation was used to test hypotheses using mixed models in SPSS. Specially, cross-classified random effects models were conducted, as multiple raters viewed multiple focal participants, making the data crossed and non-independent, preventing usage of more common general linear model techniques. The first set of models examined the effect of different labels on first impressions, where diagnostic label (e.g. any label or "no diagnosis"), type of autism descriptor (e.g. Asperger's, autism, autistic), and the interaction of the two were entered as factors to predict each first impression item. These tests yield *F* tests of main effects and the interaction which can be followed up by examining differences between label conditions for diagnostic labels and Bonferroni corrected pairwise comparisons for autism descriptors. MLM also yields estimates of partitioned variance; thus, to parse variability in first impressions due to the stimulus participant, rater participant, or the residual variance left over, random effects for the model were estimated with variance components covariance structure.

We then examined how rater autism knowledge, stigma toward autism as assessed by the Social Distance Scale, and contact with autism influenced first impressions formed across conditions. We again used cross-classified random effects models, now including continuous predictors (i.e. autism knowledge, level of contact, and stigma), covariates (i.e. rater age and gender), and interaction terms for each continuous predictor with each labeling condition. Both labeling conditions were effects coded, and all continuous predictors were grand mean centered. These results yield *F* tests of main effects and unstandardized regression coefficients for the effect of each predictor on each first impression item. Significant two-way interactions of rater characteristics with both label conditions were followed up by dummy coding each condition and estimating simple slopes.

Effect size was estimated using standardized regression coefficients. Standardized coefficients were computed for main effects of continuous variables and in simple slopes analyses by computing z-scores for outcomes and continuous predictors and using these variables in analyses. An estimate of effect size was also computed for categorical variables by regressing standardized outcomes on effects coded predictors. Specifically, diagnostic label and gender were effects coded with values 0.5 and −0.5 to obtain an estimate of the difference between any label compared to no label and males compared to females, respectively. Effects coding of descriptor labels allowed for estimates of the effect of each label on standardized outcomes compared to the average effect of the descriptor labels.

Results

Effect of autism descriptors and diagnostic labels

Correlations between the 10 first impression items and rater age, autism awareness, stigma, and level of contact with autism are displayed in Table 1 along with means and standard deviations.

We entered autism descriptor label, diagnostic label, and the interaction term as factors into separate cross-classified random effects models to predict each first impression. Significance tests of effects of labels on first impressions are displayed in Table 2. Having any diagnostic label was related to better impressions compared to "no diagnosis" ($ps < 0.001$), but there were no effects of autism descriptors ($ps > 0.10$). Means and standard errors for first impressions across label types are displayed in Figure 1.

These models also estimated partitioned variance (Table 2). On 9 of 10 items, variability was driven more by raters than stimulus participants. For the "intentions to interact" items, 29%–45% of the variance was accounted for by raters compared to 4%–8% accounted for by stimulus participants. In all, 95% confidence intervals (CIs) for rater and stimulus participant variance on these items did not overlap, indicating statistically meaningful differences. Large residual variance (e.g. the unique combination of raters and stimulus participants, and error variance) suggested that idiosyncratic rater responses to individual stimulus participants, along with other unexplained factors, also contributed to ratings. Stimulus participants only accounted for more variance than raters on "awkwardness," suggesting that this judgment was driven more by characteristics of the autistic adults than of the raters.

Effect of rater characteristics and labels

We next entered rater characteristics, labeling conditions, and the interaction of rater characteristics and labels as continuous predictors into each cross-classified random effects model to predict each first impression item. Significance tests for all fixed effects in the models are displayed in Table 3. Standardized regression coefficients capturing the effect size of the main effects of gender, diagnostic label, and descriptor labels are displayed in Table 4. Standardized and unstandardized regression coefficients for the main effects of age, level of contact, autism awareness, and social distance are displayed in Table 5. Controlling for rater demographics and characteristics, there was still an effect of diagnostic label on all items ($ps < 0.01$) except dominance ($p = 0.15$) and no effect of autism descriptor ($ps > 0.31$). There were also main effects of raters' characteristics (Table 5). Higher stigma (i.e. higher scores on the Social Distance Scale) predicted less favorable ratings across all items ($ps < 0.03$) except dominance ($p = 0.15$).

Table 1. Zero-order correlations, means, and standard deviations of first impressions, level of contact, autism awareness, social distance, age, and IQ.

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Awkward_R	1														
2	Attractive	0.234**	1													
3	Trustworthy	0.200**	0.193**	1												
4	Dominance	0.113**	0.026**	−0.201**	1											
5	Likable	0.332**	0.322**	0.480**	−0.226**	1										
6	Smart	0.246**	0.291**	0.310**	−0.008	0.326**	1									
7	Live near	0.192**	0.176**	0.311**	−0.258**	0.379**	0.238**	1								
8	Hangout	0.311**	0.444**	0.339**	−0.079**	0.497**	0.332**	0.281**	1							
9	Sit near	0.261**	0.217**	0.322**	−0.238**	0.410**	0.273**	0.592**	0.342**	1						
10	Conversation	0.309**	0.382**	0.358**	−0.112**	0.522**	0.313**	0.297**	0.630**	0.416**	1					
11	Level of contact	−0.046**	0.017	0.031**	0.027**	0.015	0.020*	−0.005	0.01	0.008	0.019	1				
12	Autism awareness	−0.030**	−0.028**	0.083**	−0.046**	0.050**	0.089**	0.189**	−0.024*	0.147**	−0.005	0.212**	1			
13	Social distance	−0.024*	−0.142**	−0.108**	0.051**	−0.157**	−0.262**	−0.207**	−0.215**	−0.205**	−0.227**	−0.116**	−0.330**	1		
14	Age	−0.034**	0.037**	−0.025*	0.022*	−0.007	0.034**	−0.038**	−0.006	−0.003	0.062**	0.018	−0.099**	−0.074**	1	
15	WRAT standard score	−0.049**	0.022*	0.037**	0.028**	−0.017	−0.016	0.122**	−0.013	0.050**	−0.017	0.021*	0.279**	−0.063**	−0.177**	
	M	2.4	2.01	2.8	1.9	2.78	2.67	3.07	2.13	3.04	2.43	6.87	12.08	11.56	21.37	108.04
	SD	0.82	0.59	0.59	0.75	0.61	0.67	0.68	0.66	0.68	0.72	2.99	5.14	3.66	4.98	8.82

WRAT: Wide Range Achievement Test; SD: standard deviation.
Awkwardness was reverse scored.
*$p < 0.05$; **$p < 0.001$.

Table 2. Fixed effects of autism descriptor label and diagnostic label on first impressions.

	Main effect of diagnostic label		Main effect of autism descriptor		Interaction		Stimulus variance	Rater variance	Residual variance
	F(1, 9571)	*p*	*F*(4, 500)	*p*	*F*(4, 9571)	*p*			
Awkward_R	73.715*	<0.001	0.157	0.96	1.953	0.10	25.90%	14.36%	59.74%
Attractive	106.796*	<0.001	0.55	0.70	1.643	0.16	13.40%	36.11%	50.49%
Trustworthy	254.382*	<0.001	0.228	0.92	1.933	0.10	10.80%	19.62%	69.57%
Dominant	11.098*	<0.001	1.689	0.15	1.849	0.12	15.52%	25.26%	59.22%
Likable	301.559*	<0.001	1.057	0.38	1.19	0.31	14.41%	16.86%	68.73%
Smart	14.366*	<0.001	1.035	0.39	1.049	0.38	6.41%	41.15%	52.44%
Live near	85.063*	<0.001	0.908	0.46	1.481	0.21	4.33%	45.38%	50.30%
Hangout	154.163*	<0.001	1.95	0.10	2.23	0.06	7.70%	29.18%	63.12%
Sit near	122.815*	<0.001	0.825	0.51	2.122	0.08	5.77%	39.92%	54.31%
Conversation	140.193*	<0.001	0.372	0.83	1.476	0.21	6.94%	36.05%	57.00%

Awkwardness was reverse scored.
*$p < 0.001$.

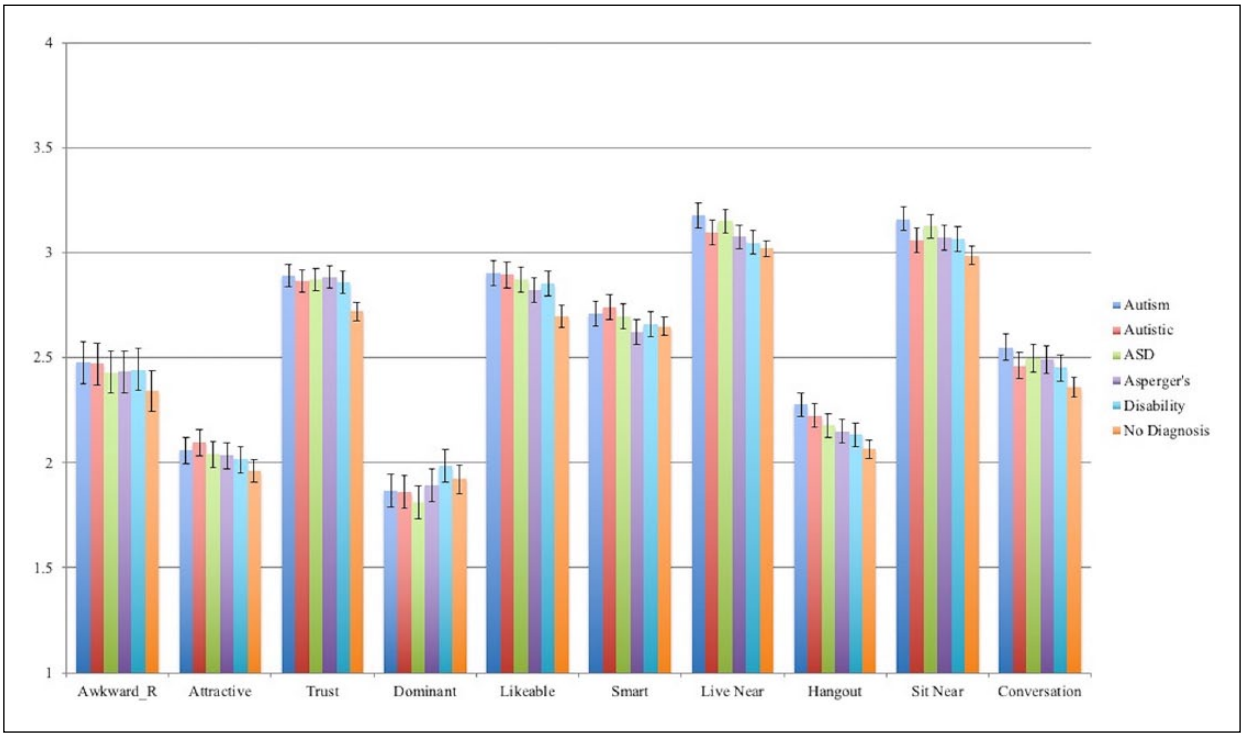


Figure 1. Mean differences in first impressions across the different label types.
Error bars indicate standard error. Awkwardness was reverse scored.

Higher autism knowledge was related to willingness to live near and comfort sitting next to the stimulus participant ($ps < 0.01$) and more negative ratings of awkwardness, attractiveness, and willingness to hang out and start a conversation ($ps < 0.04$). Autism contact was only related to higher dominance and awkwardness ratings ($ps < 0.05$).

Main effects were qualified by interactions with diagnostic labels ($ps < 0.03$). The negative main effect of stigma on first impressions was magnified for labeling conditions, such that raters with greater levels of stigma rated stimuli less favorably on intelligence (any descriptor: $b = -0.05$, $\beta = -0.26$, $SE = 0.006$, $p < 0.01$; no diagnosis: $b = -0.04$, $\beta = -0.23$, $SE = 0.006$, $p < 0.001$), willingness to hang out (any descriptor: $b = -0.05$, $\beta = -0.29$, $SE = 0.005$, $p < 0.01$; no diagnosis: $b = -0.04$, $\beta = -0.22$, $SE = 0.005$, $p < 0.001$), and start a conversation (any descriptor: $b = -0.06$, $\beta = -0.28$, $SE = 0.005$, $p < 0.01$; no diagnosis: $b = -0.05$, $\beta = -0.23$, $SE = 0.005$, $p < 0.001$) ratings

Table 3. Fixed effects of descriptor label, diagnostic label, and rater characteristics on first impressions.

	df	Awkward_R		Attractive		Trust		Dominance		Likable		Smart		Live near		Hangout		Sit near		Conversation	
		F	*p*	*F*	*p*	*F*	*p*	*F*	*p*	*F*	*p*	*F*	*p*	*F*	*p*	*F*	*p*	*F*	*p*	*F*	*p*
Descriptor label	(4, 475)	0.243	0.914	0.619	0.649	0.221	0.927	0.713	0.583	0.649	0.628	0.432	0.786	1.212	0.305	0.777	0.54	1.241	0.293	1.213	0.305
Diagnostic label	(1, 9566)	43.71*	<0.001	66.491*	<0.001	184.336*	<0.001	2.09	0.148	189.317*	<0.001	6.741*	0.009	42.783*	<0.001	92.504*	<0.001	72.081*	<0.001	86.3*	<0.001
Gender	(1, 475)	0.555	0.457	5.502*	0.019	3.55	0.06	2.981	0.085	1.608	0.205	6.554*	0.011	1.398	0.238	4.869*	0.028	0.005	0.942	3.225	0.073
Level of contact	(1, 475)	4.73*	0.03	0.74	0.39	0.31	0.578	4.024*	0.045	0.031	0.86	0.149	0.7	3.004	0.084	0.569	0.451	1.209	0.272	0.211	0.646
Autism awareness	(1, 475)	4.421*	0.036	8.573*	0.004	3.211	0.074	2.22	0.137	0.136	0.713	0.004	0.948	17.178*	<0.001	21.074*	<0.001	8.024*	0.005	8.244*	0.004
Social distance	(1, 475)	4.835*	0.028	33.767*	<0.001	12.741*	<0.001	2.065	0.151	49.934*	<0.001	65.937*	<0.001	22.283*	<0.001	99.16*	<0.001	32.648*	<0.001	80.348*	<0.001
Age	(1, 475)	5.561*	0.019	0.952	0.33	1.366	0.243	1.627	0.203	1.099	0.295	0.039	0.843	0.118	0.731	0.855	0.356	0.281	0.597	2.177	0.141
Diagnostic label × Descriptor label	(4, 9566)	1.908	0.106	1.534	0.189	1.843	0.118	1.837	0.119	1.107	0.351	0.798	0.526	1.279	0.276	1.678	0.152	1.76	0.134	0.812	0.518
Descriptor label × Level of contact	(4, 475)	0.261	0.903	1.583	0.177	2.342	0.054	0.174	0.952	0.389	0.816	2.305	0.057	2.925*	0.021	0.711	0.584	1.367	0.244	1.088	0.362
Descriptor label × Autism awareness	(4, 475)	1.278	0.278	0.765	0.548	0.544	0.704	1.592	0.175	0.441	0.779	0.649	0.628	0.301	0.877	0.165	0.956	1.516	0.196	0.3	0.878
Descriptor label × Social distance	(4, 475)	0.786	0.535	1.779	0.132	2.043	0.087	0.333	0.856	0.802	0.524	1.931	0.104	3.088*	0.016	0.471	0.757	2.295	0.058	0.811	0.519
Descriptor label × Gender	(4, 475)	0.274	0.895	0.966	0.426	0.317	0.867	1.236	0.295	0.668	0.614	0.751	0.558	1.048	0.382	1.56	0.184	0.452	0.771	1.307	0.266
Descriptor label × Age	(4, 475)	1.718	0.145	1.123	0.345	1.85	0.118	0.617	0.651	0.507	0.73	0.616	0.652	1.323	0.26	0.595	0.666	0.631	0.641	0.06	0.993
Diagnostic label × Level of contact	(1, 9566)	0.559	0.454	0.244	0.622	0.237	0.627	0.289	0.591	0.526	0.469	0.034	0.855	0.817	0.366	0.041	0.839	0.083	0.773	0.506	0.477
Diagnostic label × Autism awareness	(1, 9566)	2.417	0.12	5.164*	0.023	3.434	0.064	0.233	0.629	14.955*	<0.001	2.009	0.156	3.404	0.065	1.615	0.204	5.461*	0.019	13.8*	<0.001
Diagnostic label × Social distance	(1, 9566)	2.137	0.144	0.81	0.368	0.247	0.619	1.462	0.227	0.118	0.731	4.514*	0.034	0.135	0.713	13.118*	<0.001	2.159	0.142	8.515*	0.004
Diagnostic label × Gender	(1, 9566)	1.283	0.257	0.983	0.321	0.1	0.752	6.264*	0.012	2.483	0.115	1.216	0.27	5.012*	0.025	2.403	0.121	2.372	0.124	1.633	0.201
Diagnostic label × Age	(1, 9566)	1.714	0.19	0.987	0.32	0.047	0.828	4.525*	0.033	0.007	0.935	0.958	0.328	1.454	0.228	5.236*	0.022	1.581	0.209	1.627	0.202

df: degrees of freedom.
Awkwardness was reverse scored.
*$p < 0.05$.

Table 4. Effect sizes for effect of gender, diagnostic label, and descriptor labels on first impressions.

	Parameter	Gender	Diagnostic label	Autism label	ASD label	Asperger's label	Autistic label	Disability label
Awkward_R	β	−0.03	0.12	−0.02	0.04	−0.01	−0.01	<0.001
	95% CI	[−0.12, 0.06]	[0.09, 0.16]	[−0.12, 0.08]	[−0.05, 0.13]	[−0.10, 0.07]	[−0.09, 0.08]	[−0.10, 0.09]
	SE	0.05	0.02	0.05	0.05	0.04	0.04	0.05
	t	−0.75	6.61	−0.44	0.95	−0.29	−0.16	−0.04
	p	0.46	<0.001	0.66	0.34	0.77	0.87	0.97
Attractive	β	0.16	0.14	0.09	−0.07	−0.01	−0.02	<0.001
	95% CI	[0.03, 0.28]	[0.10, 0.17]	[−0.05, 0.23]	[−0.20, 0.05]	[−0.13, 0.11]	[−0.14, 0.10]	[−0.13, 0.14]
	SE	0.07	0.02	0.07	0.06	0.06	0.06	0.07
	T	2.35	8.15	1.33	−1.15	−0.09	−0.30	0.06
	p	0.02	<0.001	0.18	0.25	0.93	0.77	0.95
Trustworthiness	β	−0.10	0.27	0.00	0.01	−0.01	−0.03	0.04
	95% CI	[−0.20, 0.00]	[0.23, 0.30]	[−0.11, 0.10]	[−0.09, 0.11]	[−0.10, 0.08]	[−0.13, 0.06]	[−0.06, 0.14]
	SE	0.05	0.02	0.06	0.05	0.05	0.05	0.05
	T	−1.88	13.58	−0.07	0.13	−0.21	−0.69	0.76
	p	0.06	<0.001	0.94	0.90	0.84	0.49	0.45
Dominant	β	0.10	−0.03	−0.02	−0.04	0.05	−0.06	0.06
	95% CI	[−0.01, 0.22]	[−0.06, 0.01]	[−0.14, 0.11]	[−0.15, 0.08]	[−0.06, 0.16]	[−0.17, 0.05]	[−0.06, 0.18]
	SE	0.06	0.02	0.06	0.06	0.05	0.06	0.06
	t	1.73	−1.45	−0.25	−0.62	0.93	−1.09	1.01
	p	0.09	0.15	0.80	0.53	0.35	0.28	0.32
Likeability	β	−0.06	0.27	0.01	−0.01	−0.05	−0.01	0.06
	95% CI	[−0.15, 0.03]	[0.23, 0.30]	[−0.09, 0.11]	[−0.10, 0.08]	[−0.14, 0.03]	[−0.10, 0.07]	[−0.04, 0.16]
	SE	0.05	0.02	0.05	0.05	0.04	0.04	0.05
	t	−1.27	13.76	0.27	−0.15	−1.24	−0.30	1.24
	p	0.21	<0.001	0.79	0.88	0.22	0.77	0.22
Smart	β	−0.17	0.04	0.05	−0.03	−0.06	0.01	0.05
	95% CI	[−0.30, 0.04]	[0.01, 0.08]	[−0.10, 0.19]	[−0.16, 0.10]	[−0.18, 0.06]	[−0.12, 0.13]	[−0.09, 0.18]
	SE	0.07	0.02	0.07	0.07	0.06	0.06	0.07
	t	−2.56	2.60	0.64	−0.50	−1.03	0.09	0.65
	p	0.01	0.01	0.52	0.62	0.30	0.93	0.51
Live near	β	−0.08	0.11	0.10	0.07	−0.09	−0.08	0.00
	95% CI	[−0.22, 0.05]	[0.08, 0.14]	[−0.05, 0.25]	[−0.07, 0.20]	[−0.22, 0.04]	[−0.21, 0.05]	[−0.14, 0.14]
	SE	0.07	0.02	0.08	0.07	0.07	0.07	0.07
	t	−1.18	6.54	1.32	1.00	−1.34	−1.24	0.01
	p	0.24	< 0.001	0.19	0.32	0.18	0.22	0.99
Hangout	β	0.13	0.18	0.10	−0.04	−0.01	−0.05	0.01
	95% CI	[0.01, 0.24]	[0.14, 0.22]	[−0.02, 0.22]	[−0.15, 0.07]	[−0.12, 0.09]	[−0.16, 0.06]	[−0.11, 0.13]
	SE	0.06	0.02	0.06	0.06	0.05	0.05	0.06
	t	2.21	9.62	1.57	−0.78	−0.24	−0.93	0.17
	p	0.03	<0.001	0.12	0.43	0.81	0.36	0.86
Sit near	β	0.00	0.15	0.08	0.06	−0.06	−0.11	0.03
	95% CI	[−0.14, 0.13]	[0.11, 0.18]	[−0.07, 0.22]	[−0.07, 0.19]	[−0.18, 0.07]	[−0.24, 0.01]	[−0.11, 0.17]
	SE	0.07	0.02	0.07	0.07	0.06	0.06	0.07
	t	−0.07	8.49	1.05	0.94	−0.92	−1.77	0.45
	p	0.94	<0.001	0.29	0.35	0.36	0.08	0.65
Conversation	β	0.11	0.17	0.06	0.00	0.07	−0.11	−0.03
	95% CI	[−0.01, 0.24]	[0.13, 0.20]	[−0.07, 0.20]	[−0.12, 0.12]	[−0.04, 0.19]	[−0.23, 0.01]	[−0.16, 0.10]
	SE	0.06	0.02	0.07	0.06	0.06	0.06	0.07
	t	1.80	9.29	0.93	−0.03	1.25	−1.82	−0.40
	p	0.07	<0.001	0.35	0.98	0.21	0.07	0.69

CI: confidence interval; SE: standard error; ASD: autism spectrum disorder.
Awkwardness was reverse scored; all predictors were coded with effects coding and were regressed on standardized outcome variables; Diagnostic Label and Gender were effects coded with values 0.5 and −0.5 to obtain an estimate of the difference between any label compared to no label and males compared to females, respectively; and standardized regression coefficients for descriptor labels represent the contrast comparing each label to the average effect of the other descriptor labels.

Table 5. Unstandardized and standardized regression coefficients for main effects of age, level of contact, autism knowledge, and social distance on first impressions.

	Age				Level of contact				Autism knowledge				Social distance			
	b	β	*SE*	*p*	*b*	β	*SE*	*p*	*b*	β	*SE*	*p*	*b*	β	*SE*	*p*
Awkward_R	−0.009*	−0.05	0.003	0.02	−0.01*	−0.04	0.005	0.030	−0.01*	−0.04	0.003	0.036	−0.01*	−0.05	0.005	0.028
Attractive	0.004	0.03	0.004	0.33	0.001	0.02	0.06	0.390	−0.01*	−0.09	0.003	0.004	−0.03*	−0.17	0.005	<0.001
Trustworthy	−0.004	−0.03	0.003	0.24	0.001	0.01	0.004	0.578	0.001	0.04	0.003	0.074	−0.01*	−0.08	0.004	<0.001
Likable	−0.003	−0.02	0.003	0.30	−0.001	−0.004	0.004	0.860	−0.001	−0.008	0.003	0.713	−0.03*	−0.15	0.004	<0.001
Dominant	−0.005	−0.04	0.004	0.20	0.01*	0.05	0.006	0.045	−0.01	−0.04	0.004	0.137	0.01	0.04	0.005	0.151
Smart	−0.001	−0.006	0.004	0.84	−0.001	−0.01	0.006	0.700	0.001	0.002	0.004	0.948	−0.04*	−0.24	0.005	<0.001
Live near	−0.002	−0.01	0.005	0.73	−0.01	−0.05	0.007	0.084	0.02*	0.13	0.004	<0.001	−0.03*	−0.15	0.006	<0.001
Hangout	−0.003	−0.03	0.004	0.36	0.001	0.02	0.005	0.451	−0.02*	−0.12	0.003	<0.001	−0.05*	−0.26	0.005	<0.001
Sit near	0.002	0.02	0.004	0.60	−0.01	−0.03	0.007	0.272	0.01*	0.09	0.004	0.005	−0.03*	−0.17	0.006	<0.001
Conversation	0.006	0.04	0.004	0.14	0.001	0.01	0.007	0.646	−0.01*	−0.08	0.004	0.004	−0.05*	−0.26	0.006	<0.001

SE: standard error.
Awkwardness was reverse scored; Models were run with all categorical variables effects coded.
*$p < 0.05$.

compared to the "no diagnosis" conditions. Stigma also interacted with autism descriptors; raters with more stigma minded living near stimulus participants labeled "autistic" ($b = -0.06$, $\beta = -0.31$, $SE = 0.01$, $p < 0.001$) and "autism" ($b = -0.04$, $\beta = -0.24$, $SE = 0.01$, $p < 0.001$).

When including a diagnostic label, the positive effects of autism knowledge on comfort sitting next to stimulus participants were stronger with any descriptor ($b = 0.014$, $\beta = 0.10$, $SE = 0.004$, $p = 0.001$) compared to "no diagnosis" ($b = 0.009$, $\beta = 0.07$, $SE = 0.004$, $p = 0.03$). More knowledge also predicted more likeability ratings with any descriptor ($b = 0.003$, $\beta = 0.03$, $SE = 0.003$, $p = 0.249$) compared to "no diagnosis" ($b = -0.005$, $\beta = -0.04$, $SE = 0.003$, $p = 0.07$). Although this interaction term was significant ($p < 0.01$), indicating a meaningful difference between conditions, the individual coefficient only trended toward significance from zero. In addition, prior negative effects of autism knowledge were attenuated for attractiveness (any descriptor: $b = -0.008$, $\beta = -0.07$, $SE = 0.004$, $p = 0.02$; no diagnosis: $b = -0.012$, $\beta = -0.1$, $SE = 0.004$, $p = 0.001$) and willingness to start a conversation (any descriptor: $b = -0.007$, $\beta = -0.05$, $SE = 0.004$, $p = 0.08$; no diagnosis: $b = -0.015$, $\beta = -0.11$, $SE = 0.004$, $p < 0.001$) when a diagnostic label was present compared to "no diagnosis."

Raters with more contact with autism were less willing to live near participants labeled with "autism" ($b = -0.04$, $\beta = -0.17$, $SE = 0.01$, $p = 0.01$), but contact interacted with no other conditions ($ps > 0.05$).

We also examined how age and gender of the raters may influence first impressions and interact with other rater characteristics. There was a main effect of gender (see Table 4) such that controlling for other variables in the model, males rated autistic adults as more attractive ($p = 0.02$) and were more willing to hang out with them than females ($p = 0.03$), but also rated them as less intelligent ($p = 0.01$). Although there was no interaction of gender with descriptive label ($ps < 0.18$), gender did interact with diagnostic label on ratings of willingness to live near stimuli ($p = 0.01$) and ratings of dominance ($p = 0.03$). Simple slopes show females were more willing than males to live near a stimulus participant when a label was provided (any label: $b = -0.041$,$\beta = -0.061$, $SE = 0.02$, $p = 0.094$; no label: $b = -0.0015$,$\beta = -0.022$, $SE = 0.02$, $p = 0.54$); however, the simple slopes analysis revealed these effects were not significantly different from zero. Males also rated stimulus participants as more dominant than females when a label was provided (any label: $b = 0.05$, $\beta = 0.07$, $SE = 0.02$, $p = 0.02$; no label: $b = 0.020$, $\beta = 0.027$, $SE = 0.02$, $p = 0.38$).

There was a main effect of age on awkwardness, with older raters rating autistic adults as more awkward (see Table 5). Age did not interact with autism-descriptor label ($p = 0.12$), but did interact with diagnosis on ratings of dominance and willingness to hang out with, such that older participants rated autistic adults with diagnostic labels as more dominant and were less willing to hang out

with autistic adults given diagnostic labels compared to the no diagnosis label condition; however, the simple slopes indicated these regression coefficients did not significantly differ from zero (dominance ratings label: $b = 0.01$, $\beta = 0.05$, $SE = 0.004$, $p = 0.07$; dominance ratings no label: $b = 0.003$, $\beta = 0.02$, $SE = 0.004$, $p = 0.52$; hangout ratings label: $b = -0.01$, $\beta = 0.04$, $SE = 0.004$, $p = 0.13$; hangout ratings no label: $b = -0.001$, $\beta = -0.007$, $SE = 0.004$, $p = 0.81$).

Discussion

This study found that variability in first impressions of autistic adults made by NT raters occurs more at the level of the rater than the autistic adult. This pattern emerged for 9 out of 10 items and was particularly large for the "intent to interact" impressions, indicating NT interest in social interaction was driven more by individual differences among the raters than among the autistic adults they were rating. Thus, the findings reported here provide evidence that social opportunities for autistic adults among NT peers may be dictated to a large degree by NT characteristics and attitudes rather than just their own self-presentation and social behavior.

The individual differences among NT raters that were associated with variability in first impressions of autistic adults included stigma-related beliefs about autism and, to a lesser degree, autism knowledge. NT raters expressing higher stigma-related beliefs offered less favorable impressions of autistic adults on all but one item, and several of these became even more negative when the diagnoses of autistic adults were made explicit. This finding suggests that awareness of a diagnosis may activate stigma-related beliefs and accentuate dissimilarity between rater and target (Komeda, 2015). Reducing autism-related stigma among NT individuals may therefore serve as one avenue to improve acceptance of autistic differences. Prior work within a university sample indicated that even brief online training can decrease stigma and increase acceptance of autistic peers (Gillespie-Lynch et al., 2015), and greater NT contact and familiarity with autistic individuals is associated with more inclusive and accepting attitudes of autism (Gardiner & Iarocci, 2014). Thus, consistent with the broader social psychological literature indicating that meaningful intergroup contact can reduce bias and prejudice (Pettigrew & Tropp, 2000), autism-related stigma among NTs may be expected to decrease with greater societal inclusion and integration of autistic people.

Meanwhile, we found that higher autism knowledge among NT raters produced mixed results when diagnoses were withheld, but these impressions improved when a diagnosis was included. Thus, consistent with prior findings (Sasson & Morrison, 2019), autism knowledge may only produce benefits when diagnoses are known, suggesting knowledge does not help detect autistic characteristics but instead may soften evaluations when a diagnosis is known. It may be the case that higher autism knowledge among NTs co-occurs with more inclusive and accepting attitudes, and thus, autism knowledge only relates to more favorable impressions when autism status is explicitly signaled. From this perspective, programs that aim exclusively to increase autism knowledge among NTs rather than also reduce stigma or increase meaningful interaction with autistic people may be limited in their benefit to the experiences of autistic people. This interpretation is consistent with Gardiner and Iarocci (2014), who found that direct contact but not autism knowledge was associated with acceptance of autistic peers among university students. Direct contact also breeds familiarity, which may increase interpretability of distinct modes of social presentation and behavior, and over time, mitigate aspects of the "double empathy problem" (Milton, 2012) of miscommunication and misunderstanding between autistic and non-autistic people.

Replicating previous studies (Matthews et al., 2015; Sasson & Morrison, 2019), a general benefit of disclosing autism status on first impressions was found in this study. However, several important qualifiers to this conclusion emerged. First, the effect does not depend upon the term used to describe autism (e.g. "autistic" vs "has autism"). The finding of a positive benefit of including any specific diagnostic term aligns with Gernsbacher et al. (2016), who reported more favorable evaluations of specific disability terms compared to general euphemisms and suggest that how autistic adults refer to their diagnosis does not affect how they are perceived. Second, the general benefit of diagnostic disclosure did not extend to NT individuals with high autism-related stigma. For them, an autism diagnosis actively *worsened* impressions, suggesting that responses to diagnostic disclosure may depend upon the beliefs and biases of those receiving the disclosure. This finding has practical implications for decisions autistic adults may make regarding disclosing their diagnosis to others. Many autistic adults are rightfully concerned whether disclosure might increase discrimination in personal and professional contexts (Campbell, 2007) and may engage in effortful and stressful strategies to mask their autistic characteristics (Hull et al., 2017). The results reported here suggest that the prior studies demonstrating a benefit of diagnostic disclosure among university students (Matthews et al., 2015; Sasson & Morrison, 2019) may have been driven in part by high levels of autism acceptance among this population (White et al., 2016). Because disclosure may produce negative effects on impressions among NTs with less accepting attitudes, decreasing stigma toward autism in the general population may improve social experiences for autistic adults.

Importantly, raters expressing stigma toward autism provided less favorable first impressions of autistic participants even when their diagnosis was withheld. This may

suggest that those who hold negative attitudes about autism are more negative in their evaluations of others generally, even in the absence of knowledge of the person's diagnostic status. However, the fact that first impressions worsened when a diagnosis was provided indicates that NT raters with high stigma toward autism (1) did not fully detect autism and evaluate the person similarly absent the diagnosis and (2) the knowledge of the person's autism contributed to the formation of negative impressions above and beyond the autistic person's characteristics alone. Thus, the discrepancy in first impressions between when a diagnosis was provided versus when it was withheld may suggest that aspects of autism remain "invisible" to NT perceivers, and awareness of the diagnosis evokes beliefs and biases that affect impression formation, either positively or negatively depending on characteristics of the rater. The majority of our sample of undergraduate raters did not express much stigma toward autism, and for them, first impressions of autistic participants improved when informed of their diagnosis. However, in a prior study, a similar undergraduate population at the same university provided more negative first impressions when individuals were labeled as having schizophrenia relative to when they were labeled as being autistic (Morrison & Sasson, 2018). Schizophrenia is a highly stigmatized condition, and this finding suggests that different conditions may evoke different assumptions and expectations that affect perceptions and subsequent social behavior.

This study has several limitations that should be kept in mind when interpreting the reported findings. First, the findings here are limited to variability in judgments when evaluating a group of largely male autistic adults without intellectual disability. Patterns in the proportions of variance accounted for by the stimulus and rater participants may have differed if a more diverse group of autistic adults participated, or if a comparison of NT adults were included as stimulus participants. Videos of NT adults were not included to maintain the focus of the study on probing the source of variability in NT first impressions of autistic adults. Subsequent research may choose to include NT videos as a comparison to assess whether the source of variability in first impressions differ for NT and ASD stimulus participants. In addition, many unmeasured individual differences among NT raters may have influenced first impressions of autistic adults. Those examined here (e.g. stigma) were selected because of their proximal relationship to attitudes about autism, but many other individual differences, like political ideology and social dominance (Samochowiec, Wänke, & Fiedler, 2010; Todorov, Said, Engell, & Oosterhof, 2008), could contribute to how NTs form impressions of autistic people.

The psychometric properties of the measures used here should also be taken into consideration. Although the autism knowledge survey showed adequate reliability, it was still much lower than that found in the original study (Gillespie-Lynch et al., 2015). In addition, the autism contact measure used did not significantly predict many first impressions and was correlated with autism knowledge and stigma only minimally. This raises questions about its validity in this context, particularly because others have found experiences with autism do predict more favorable outcomes (Gardiner & Iarocci, 2014). Specific contact with autism as 1 of 12 broad relationships may not adequately capture the effect of sustained and quality interactions with autistic people. In addition, the nature of the regression analyses conducted here did not allow for the correction of multiple comparisons. Although we intentionally recruited and tested a large sample of raters to reduce the likelihood of Type I errors, readers should take this under consideration when interpreting the reported results.

Furthermore, although we found evidence for rater characteristics impacting impressions, the nature of our rater sample may have also influenced how impressions were formed, and thus, may limit generalizability to other samples and populations. First, only university students currently enrolled in a psychology course were included as raters. These raters were similar in age to the stimulus participants, and thus, were intended to serve as virtual "peers" of those they were rating. However, findings may have differed among raters who were older, less familiar with psychology, or from non-college populations. Indeed, we found some hints of these differences in our findings: older NT raters not only rated autistic adults as more awkward than younger NTs, but also expressed a greater reluctance to interact with them. Such results may suggest that younger NTs harbor more inclusive attitudes about autism than older NTs. Second, reported results were derived from a predominately female sample of raters and findings may have differed with a larger proportion of males. Although our rater sample was large enough to detect some gender differences among raters, future work is encouraged to explore these effects more robustly, and expand beyond them to examine if male and female autistic adults are be perceived differently and whether this depends upon the gender of the rater.

These limitations notwithstanding, findings suggest that first impressions of autistic adults may be driven in large part by NT individuals' attitudes toward autism. Variability in impression ratings were driven more by characteristics of the NT raters than by those of the autistic adults, particularly for responses about intentions to interact with the autistic adult. In particular, variability among NTs in autism-related stigma and autism knowledge were predictive of their first impressions of autistic adults. Collectively, these findings suggest that reducing autism-related stigma among NT peers may help improve personal and professional social opportunities for autistic adults. Because NT attitudes and knowledge about autism may be more malleable characteristics than the behavioral

and social differences of autistic adults—which in this study did not account for significant variability in NT first impression formation—treatment programs focusing on increasing NT acceptance and accommodation of autistic differences may hold significant promise for improving the social experiences of cognitively able autistic adults, especially within NT environments. Treatment studies testing this proposition are recommended.

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Note

1. Use of identity-first language was adopted because of the expressed preference of the adult autism community (Kenny et al., 2016).

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